LAB 06 REPORT

Fullstack Skeleton using Node.js and GitHub

Course: Web Technologies

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Contents

1 Introduction **3**

2 Project Structure **4**

3 Backend Implementation **5**
3.1 Installation of Dependencies 5
3.2 Server Execution 5

4 MongoDB Integration **7**
4.1 MongoDB Atlas Dashboard 7
4.2 Successful Database Connection 7

5 Git and GitHub Integration **9**
5.1 Git Initialization 9
5.2 Git Commit 9
5.3 Push to GitHub 9

6 Conclusion **11**

List of Figures

3.1 Installation of Required Node Modules 5
3.2 Backend Server Running Successfully 6

4.1 MongoDB Atlas Dashboard 7
4.2 MongoDB Successfully Connected 8

5.1 Git Initialization and Add Commands 9
5.2 Git Commit Execution 10
5.3 Project Successfully Pushed to GitHub 10

Chapter 1

Introduction

This lab focuses on building a fullstack skeleton project using Node.js and Express. The backend server was configured and connected with MongoDB Atlas. The project was version controlled using Git and pushed to GitHub.

Chapter 2

Project Structure

The project directory was organized as follows:

```
lab06-node-fullstack-skeleton/
|-- backend/
|   |-- src/server.js
|-- frontend/
|-- .gitignore
|-- README.md
```

Chapter 3

Backend Implementation

3.1 Installation of Dependencies

```
(.venv) PS D:\PycharmProject\Web_Technologies\lab06-node-fullstack-skeleton> cd backend
(.venv) PS D:\PycharmProject\Web_Technologies\lab06-node-fullstack-skeleton\backend> npm init -y
Wrote to D:\PycharmProject\Web_Technologies\lab06-node-fullstack-skeleton\backend\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}


(.venv) PS D:\PycharmProject\Web_Technologies\lab06-node-fullstack-skeleton\backend> npm install express cors dotenv

added 68 packages, and audited 69 packages in 2s

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
(.venv) PS D:\PycharmProject\Web_Technologies\lab06-node-fullstack-skeleton\backend> npm install -D nodemon

added 26 packages, and audited 95 packages in 2s

29 packages are looking for funding
  run `npm fund` for details
```

Figure 3.1: Installation of Required Node Modules

The required dependencies were installed successfully using npm install.

3.2 Server Execution

The backend server started successfully and was accessible at localhost.

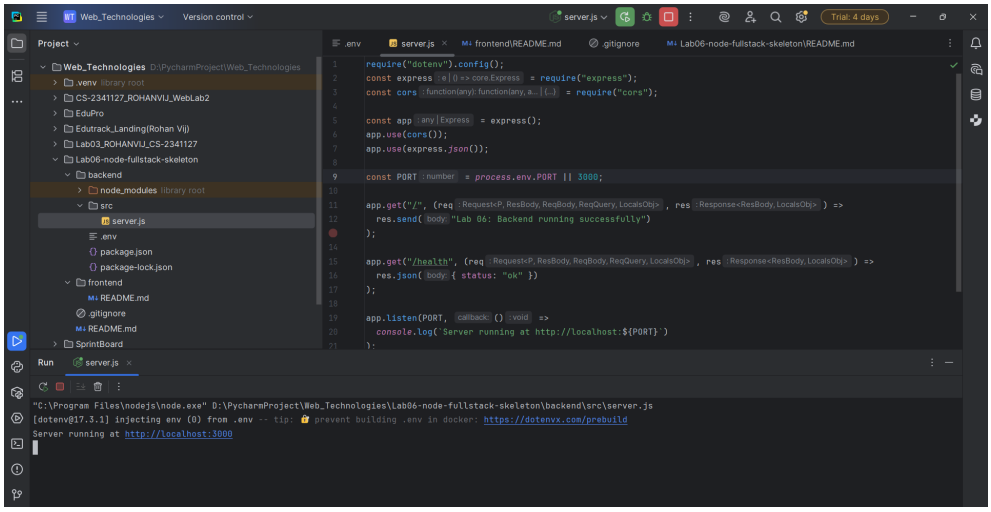

Figure 3.2: Backend Server Running Successfully

Chapter 4

MongoDB Integration

4.1 MongoDB Atlas Dashboard

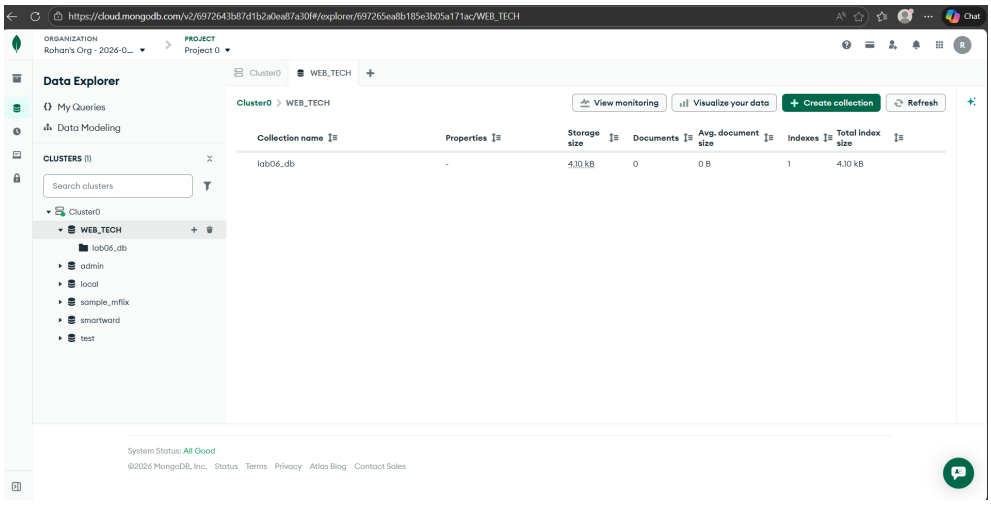

Figure 4.1: MongoDB Atlas Dashboard

The MongoDB Atlas dashboard confirms that the cluster is active.

4.2 Successful Database Connection

The console output confirms that the database connection was established successfully.

```
Terminal    Local

(.venv) PS D:\PycharmProject\Web_Technologies\Lab06-node-fullstack-skeleton\backend> npm install mongoose

added 17 packages, and audited 112 packages in 4s

30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
(.venv) PS D:\PycharmProject\Web_Technologies\Lab06-node-fullstack-skeleton\backend> npm run dev

> backend@1.0.0 dev
> nodemon src/server.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node src/server.js`
[dotenv@17.3.1] injecting env (2) from .env -- tip: 🔑prevent building .env in docker: https://dotenvx.com/prebuild
Server running at http://localhost:3000
MongoDB Connected
```

Figure 4.2: MongoDB Successfully Connected

Chapter 5

Git and GitHub Integration

5.1 Git Initialization

```
 create mode 100644 EduPro/Screenshot 2026-02-10 141156.png
 create mode 100644 EduPro/index.html
 create mode 100644 EduPro/styles.css
 create mode 100644 Edutrack_Landing(Rohan Vij)/Screenshot 2026-02-16 155031.png
 create mode 100644 Edutrack_Landing(Rohan Vij)/Screenshot 2026-02-16 155128.png
 create mode 100644 Edutrack_Landing(Rohan Vij)/css/styles.css
 create mode 100644 Edutrack_Landing(Rohan Vij)/index.html
 create mode 100644 Edutrack_Landing(Rohan Vij)/js/app.js
 create mode 100644 Lab03_ROHANVIJ_CS-2341127/assets/club.jpg
 create mode 100644 Lab03_ROHANVIJ_CS-2341127/assets/logo.png
 create mode 100644 Lab03_ROHANVIJ_CS-2341127/index.html
 create mode 100644 Lab03_ROHANVIJ_CS-2341127/style.css
 create mode 100644 Lab06-node-fullstack-skeleton/.gitignore
 create mode 100644 Lab06-node-fullstack-skeleton/README.md
 create mode 100644 Lab06-node-fullstack-skeleton/backend/package-lock.json
 create mode 100644 Lab06-node-fullstack-skeleton/backend/package.json
 create mode 100644 Lab06-node-fullstack-skeleton/backend/src/server.js
 create mode 100644 Lab06-node-fullstack-skeleton/frontend/README.md
 create mode 100644 SprintBoard/Screenshot 2026-02-21 152226.png
 create mode 100644 SprintBoard/Screenshot 2026-02-21 152620.png
 create mode 100644 SprintBoard/index.html
 create mode 100644 SprintBoard/script.js
 create mode 100644 SprintBoard/style.css
(.venv) PS D:\PycharmProject\Web_Technologies> git branch -M main
(.venv) PS D:\PycharmProject\Web_Technologies> git remote remove origin
(.venv) PS D:\PycharmProject\Web_Technologies> git remote add origin https://github.com/ROHANVIJ2006/Web_Technologies.git
(.venv) PS D:\PycharmProject\Web_Technologies> git push -u origin main
Enumerating objects: 52, done.
Counting objects: 100% (52/52), done.
Delta compression using up to 28 threads
Compressing objects: 100% (46/46), done.
Writing objects: 100% (52/52), 5.59 MiB | 1.91 MiB/s, done.
Total 52 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ROHANVIJ2006/Web_Technologies.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(.venv) PS D:\PycharmProject\Web_Technologies>
```

Figure 5.1: Git Initialization and Add Commands

The repository was initialized and files were staged.

5.2 Git Commit

The staged files were committed successfully.

5.3 Push to GitHub

The project was successfully pushed to the GitHub repository.

```
(.venv) PS D:\PycharmProject\Web_Technologies> git add .
warning: in the working copy of '.idea/inspectionProfiles/profiles_settings.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Lab06-node-fullstack-skeleton/backend/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Lab06-node-fullstack-skeleton/backend/package.json', LF will be replaced by CRLF the next time Git touches it
(.venv) PS D:\PycharmProject\Web_Technologies> git commit -m "Initial commit with all labs and proper gitignore"
[main (root-commit) 6fc1213] Initial commit with all labs and proper gitignore
 36 files changed, 3882 insertions(+)
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/Web_Technologies.iml
 create mode 100644 .idea/inspectionProfiles/profiles_settings.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 .idea/modules.xml
 create mode 100644 .idea/vcs.xml
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/about.html
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/assets/banner.jpg
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/assets/classroom.jpg
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/assets/logo.jpg
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/contact.html
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/index.html
 create mode 100644 CS-2341127_ROHANVIJ_WebLab2/services.html
 create mode 100644 EduPro/Screenshot 2026-02-10 141156.png
 create mode 100644 EduPro/index.html
 create mode 100644 EduPro/styles.css
 create mode 100644 Edutrack_Landing(Rohan Vij)/Screenshot 2026-02-16 155031.png
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 create mode 100644 Lab06-node-fullstack-skeleton/.gitignore
 create mode 100644 Lab06-node-fullstack-skeleton/README.md
 create mode 100644 Lab06-node-fullstack-skeleton/backend/package-lock.json
 create mode 100644 Lab06-node-fullstack-skeleton/backend/package.json
 create mode 100644 Lab06-node-fullstack-skeleton/backend/src/server.js
```

Figure 5.2: Git Commit Execution

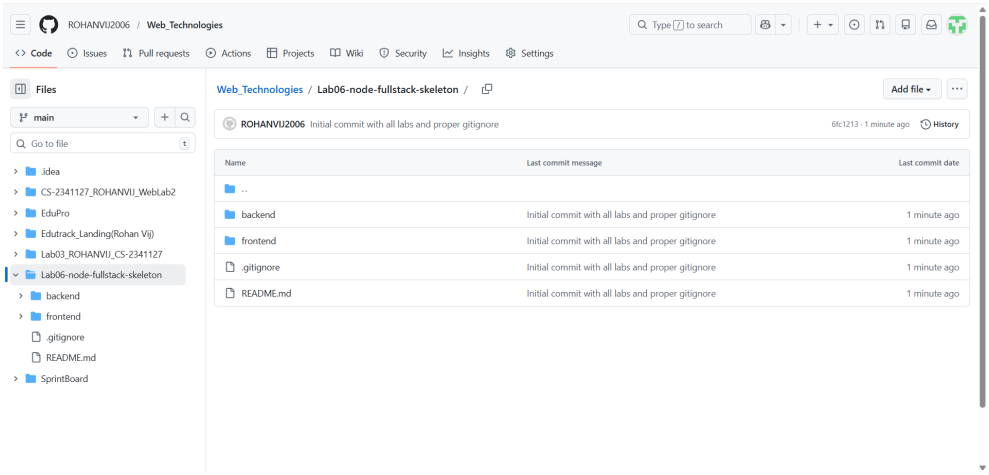

Figure 5.3: Project Successfully Pushed to GitHub

Chapter 6

Conclusion

The fullstack skeleton project was implemented successfully with backend configuration, MongoDB integration, and Git version control. All objectives of the lab were achieved.